

Reshma Katharin Biju

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SUMMARY

Biomedical engineer with experience in EEG/EMG signal processing, neural decoding, and ML-based motor classification advancing non-invasive BCI research. Hands-on background in GCP/HIPAA/GLP-compliant environments across neuroscience, pharmaceutical DSP, and biomedical device evaluation. Proven leader serving as BME Graduate Student Association President at the University at Buffalo.

EDUCATION

M.S. Biomedical Engineering (STEM), Aug 2024 – May 2026 | Buffalo, NY
University at Buffalo, State University of New York (CGPA: 3.87/4.00)

B.Tech Biomedical Engineering, Aug 2019 – May 2023 | India
MIT School of Bioengineering Sciences & Research; MIT ADT University (CGPA: 3.44/4.00)

EXPERIENCE

Graduate Researcher, University at Buffalo Aug 2024 – May 2026 | Buffalo, NY

- Developed EEG-to-EMG neural decoding models using NLARX and Fourier-based feature extraction, achieving up to 89% prediction accuracy and advancing non-invasive motor-intention estimation for brain-computer interface and neurorehabilitation applications.
- Built MATLAB-based multimodal EEG, EMG, TMS, and tDCS analysis pipelines, improving signal-processing efficiency and generating reproducible datasets that achieved ~83% motor-task classification accuracy for neuroscience research.
- Supported IRB-approved neuroscience studies involving 10+ participants and 20+ trials under GCP, HIPAA, and 45 CFR 46 requirements, ensuring regulatory compliance, participant safety, and high-quality clinical research data acquisition during brain stimulation and MRI-guided neuronavigation procedures.

Research Intern - Molecular Biology, Sun Pharma Advanced Research Company Jan 2023 – Jul 2023 | India

- Executed 300+ ALC1 protein purification runs, achieving 80–90% purity (SDS-PAGE/HPLC) and enabling cancer target characterization.
- Optimized lysis and chromatography conditions with Biologics/QC teams, improving yield consistency and reducing batch variability.
- Maintained GLP documentation and cross-functional coordination across R&D, QC, and manufacturing, ensuring regulatory traceability and smooth tech transfer.

Research Intern, Poultry Diagnostics and Research Centre Jun 2022 – Aug 2022 | India

- Supported diagnostic workflows across virology, microbiology, molecular biology, serology, and post-mortem departments, gaining hands-on experience in pathogen detection and sample processing.
- Assisted in molecular and QC procedures while maintaining SOP-compliant documentation and records for diagnostic operations.

Assistant Biomedical Engineer Intern, Enduro I Pvt. Ltd. Nov 2021 – Jan 2022 | India

- Interviewed 5+ amputees, translating feedback into reports informing prosthetic design and mechanical requirements in a startup.
- Researched prosthetic technologies and documented usability gaps to support cross-functional product development discussions.

SKILLS

Programming: MATLAB, Python, Scikit-learn, biomedical signal processing, CNN, Random Forest, SVM, kNN, LDA, NLARX modeling
Clinical Research: Human Subjects Research, IRB Compliance, GCP, HIPAA, FDA Regulations, 45 CFR 46, Belmont Report, Informed Consent, Protocol Adherence, Participant Coordination, Clinical Data Collection, GLP, Audit Readiness
Medical Equipment: EEG, EMG (Delsys Trigno), TMS (Magstim), tDCS (NeuroConn), MRI-guided neuronavigation (Brainsight), LabChart
Laboratory Techniques: SDS-PAGE, Western Blotting, ELISA, PCR, chromatography, recombinant protein purification, NanoDrop, UV-Vis spectroscopy,

LEADERSHIP

President, BME Graduate Student Association Aug 2025 – Jun 2026 | Buffalo, NY

- Led operations and communications for 100+ graduate students, organizing interdisciplinary networking events with faculty, researchers, industry partners including J&J; managed and oversaw \$2,746 annual budget with financial planning and accountability.
- Led organization to first-ever “Student Organization Collaboration of the Year” award via interdisciplinary initiatives and partnerships.

CERTIFICATES

Biomedical Research Investigators ☞, Biomedical Responsible Conduct of Research Course ☞, CITI Good Clinical Practice Course ☞, Health Information Privacy and Security (HIPS/HIPAA) ☞, Conflicts of Interest ☞, Lean Six Sigma White Belt Certification ☞

PROJECTS

Bionic Wrist for the Visually Impaired Amputees to read Braille, University at Buffalo Aug 2024 – Dec 2024

- Academic biomedical engineering project designing a bionic wrist for Braille interaction using EMG motion tracking, tactile sensors, and haptic feedback; modeled with 3 EMG sensors and 6 actuators, with simulation-based analysis estimating ~40% improvement in responsiveness and comfort under defined assumptions.

Breast Cancer Detection Using Deep Neural Network, University at Buffalo Aug 2024 – Dec 2024

- 91.79% AUROC, 83.08% accuracy breast cancer DNN model in Python, with imbalance handling, MCCV validation, and overfitting control.
- Improved model robustness via weighted loss, early stopping, and hyperparameter tuning to reduce bias and prediction error.